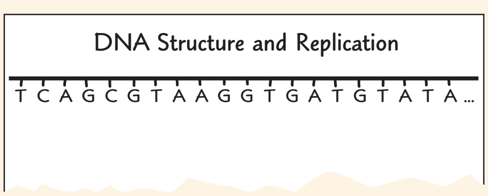
16 The Molecular Basis of Inheritance

KEY CONCEPTS

Study Tip

Draw DNA replication: Make up a single-stranded DNA sequence that is 40 nucleotides in length. As you go through Concepts 16.1 and 16.2, use your sequence to draw a double-stranded DNA molecule, labeling the ends. Then diagram the process of replication of your DNA molecule.



Go to Mastering Biology

For Students (in eText and Study Area)
- Get Ready for Chapter 16
- Figure 16.15 Walkthrough: Addition of a Nucleotide to a DNA Strand
- BioFlix® Animation: DNA Replication

For Instructors to Assign (in Item Library)
- Activity: The Hershey-Chase Experiment
- Tutorial: Visualizing DNA

Ready-to-Go Teaching Module (in Instructor Resources)
- DNA Replication (Concept 16.2)

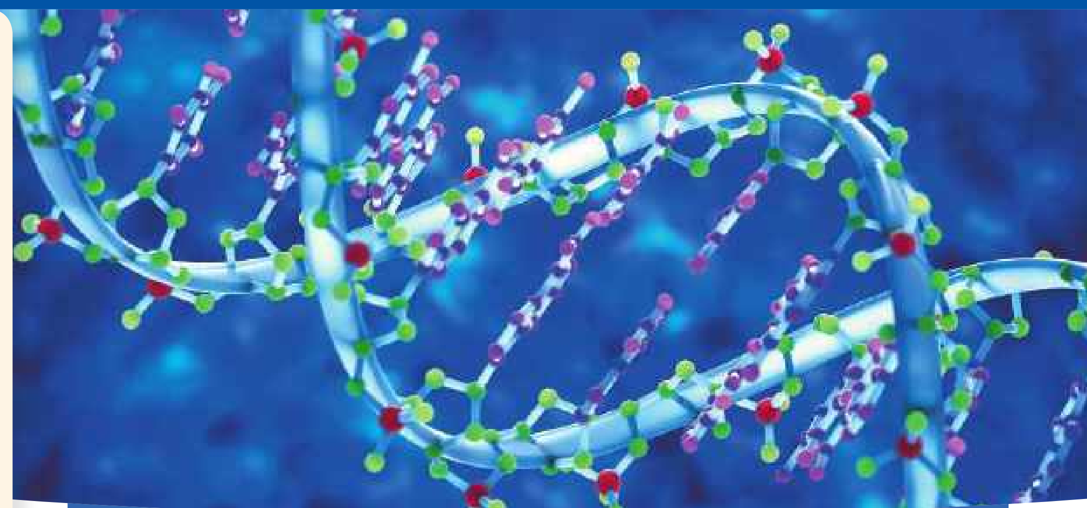


Figure 16.1 The elegant double-helical structure of deoxyribonucleic acid (DNA) shook the scientific world when it was proposed in April 1953 by James Watson and Francis Crick. The DNA you inherited from your parents contains all your genes—your genetic information.

How does DNA replication transmit genetic information?

DNA replication allows genetic information to be inherited from a parent cell to daughter cells (by mitosis) and from generation to generation (starting with meiosis).